

Daniel J. B. Smith

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EDUCATION

- University of Chicago** May 2019 – Dec 2022
Ph.D. in Evolutionary Biology
Thesis: Density-Dependent Interactions in Spatially Structured Communities
Advisor: Dr. J. Timothy Wootton
- University of Chicago** Sep 2017 – May 2019
Master of Science in Evolutionary Biology
- University of Washington** Sep 2016 – Jun 2017
Non-matriculated student
Advanced mathematics and statistics
- University of California, Los Angeles** Sep 2012 – Jun 2016
B.S. in Ecology, Behavior, and Evolution
Highest departmental honors
Senior Thesis: Toward a Mechanistic Understanding of Thermal Niches

ACADEMIC APPOINTMENTS

- The University of Texas at Austin** Sep 2024 – Present
Peter O'Donnell Jr. Postdoctoral Fellow
Advisor: Dr. Annette Ostling
Role: using data and theory to understand the effects of pathogens in tropical forest communities
- The University of Arizona** Nov 2022 – Sep 2024
Postdoctoral Researcher
Advisor: Dr. Joanna Masel
Role: building mathematical models in ecology and population genetics

PUBLICATIONS

REFEREED JOURNAL ARTICLES

* indicates authors contributed equally

- Smith, Daniel JB**. "Janzen-Connell effects and habitat-induced aggregation synergistically promote species coexistence." *The American Naturalist* (2026). [doi:10.1086/741911](https://doi.org/10.1086/741911)
- Smith, Daniel JB**, Doulcier, G., Bourrat, P., Takacs, P., and Masel, J. "Why there are so many definitions of fitness in models." *Genetics* (2026). [doi:10.1093/genetics/iyag090](https://doi.org/10.1093/genetics/iyag090)
- Magee*, L. J., **Smith*, Daniel JB**, et al. "Memories of trees past: coexistence implications of legacy conspecific density dependence." *Ecology Letters* 28.10 (2025): e70197. [doi:10.1111/ele.70197](https://doi.org/10.1111/ele.70197)
- Smith, Daniel JB** and Masel, J. "A Mechanistically Integrated Model of Exploitative and Interference Competition over a Single Resource Produces Widespread Coexistence." *The American Naturalist* 206.5 (2025): E78–E94. [doi:10.1086/737628](https://doi.org/10.1086/737628)
- LaManna, J. A., ..., **Smith, Daniel JB**, ..., Delavaux, C. "Consequences of local conspecific density effects for plant diversity and community dynamics." *Ecology Letters* 27.9 (2024): e14506. [doi:10.1111/ele.14506](https://doi.org/10.1111/ele.14506)

Magee, L. J., LaManna, J. A., Wolf, A. T., Howe, R. W., Lu, Y., Valle, D., **Smith, Daniel JB**, Bagchi, R., Bauman, D., and Johnson, D. J. "The unexpected influence of legacy conspecific density dependence." *Ecology Letters* 27.6 (2024): e14449. [doi:10.1111/ele.14449](https://doi.org/10.1111/ele.14449)

Smith, Daniel JB. "The functional form of specialised predation affects whether Janzen–Connell effects can prevent competitive exclusion." *Ecology Letters* 25.6 (2022): 1458–1470. [doi:10.1111/ele.14014](https://doi.org/10.1111/ele.14014)

Smith, Daniel JB and Amarasekare, P. "Toward a mechanistic understanding of thermal niche partitioning." *The American Naturalist* 191.3 (2018): E57–E75. [doi:10.1086/695805](https://doi.org/10.1086/695805)

IN REVIEW / SUBMITTED

Smith, Daniel JB, Forrister, D., Sedio, B. "What generates variation in conspecific negative density-dependence? A demographic perspective." *bioRxiv*. [doi:10.64898/2026.07.21.739908](https://doi.org/10.64898/2026.07.21.739908)

Smith, Daniel JB. "Life cycle structure and density dependence regulate selection on strategies generating individual heterogeneity." *In review. Manuscript available upon request.*

Gao, J., **Smith, Daniel JB**, et al. "Ontogenetic tradeoffs in conspecific negative density dependence explain coexistence of mycorrhizal strategies in a subtropical forest." *In review. Manuscript available upon request.*

Magee, L. J., Chen, Y., Worthy, S. J., Datta, D., Cox, A., **Smith, Daniel JB**, Rubio, V. E., and Swenson, N. G. "Functional gene expression and resource-dependent growth of trees highlight the mechanics of density dependent interactions." *Submitted. Manuscript available upon request.*

Smith, Daniel JB and Amarasekare, P. "Elucidating the selective mechanisms underpinning a global pattern of reproductive thermal optima in terrestrial arthropods." *In Submission. Manuscript available upon request.*

IN PREPARATION

Smith, Daniel JB, Staggs, Jon. "Asymmetric predator satiation reverses coexistence outcomes under masting and Janzen-Connell effects." *Manuscript available upon request.*

Smith, Daniel JB, Forrister, D., Magee, L. J., Sedio, B., Ostling, A. "Conspecific and chemical density legacies persist for years, shaping seedling survival." *Manuscript available upon request.*

Krishnadas, M., ..., **Smith, Daniel JB**, et al. "Conspecific density dependence in plant communities: a theory-based toolkit for empirical studies." *Manuscript available upon request.*

Van Cleve, J., Weinrich, D., **Smith, Daniel JB**, Masel, J. "Q&A: Genetic Lineages." *In submission to Genetics. Manuscript available upon request.*

ACADEMIC TALKS

INVITED TALKS

Smith, Daniel JB. "Janzen-Connell effects and habitat-induced aggregation synergistically promote species coexistence." *FOREST Theory and Empiricism (FORTE) Group Presentation, June 2026.*

Smith, Daniel JB. "A Unified Framework for Interference and Exploitative Competition." *Wolfram Summer Ecology Colloquium, August 2024. [Recording]*

Smith, Daniel JB. "A Unified Framework for Interference and Exploitative Competition." *Inaugural Meeting of the Society for Modeling and Theory in Population Biology, Banff International Research Station, Alberta, Canada, May 2024. [Recording]*

Smith, Daniel JB and Masel, J. "The Interaction Between Exploitative and Interference Competition: Limits to R* Theory." *John Templeton Foundation Early Careers Conference, Atlanta, GA, January 2024.*

Smith, Daniel JB. "The Functional Form of Specialized Predation Affects Whether Janzen-Connell Effects Can Prevent Competitive Exclusion." *Smithsonian Tropical Research Institute Virtual Science Seminar, June 2024. [Recording]*

CONTRIBUTED TALKS

- Smith, Daniel JB.** "Janzen-Connell effects and habitat-induced aggregation synergistically promote species coexistence." *Ecological Society of America Annual Meeting*, Salt Lake City, UT, 2026.
- Smith, Daniel JB** and Ostling, A. "How the abiotic environment and trait variation shape CNDD: a theoretical perspective." *61st Annual Meeting of the Association for Tropical Biology and Conservation*, Oaxaca, Mexico, 2025.
- Muller-Landau, H., ..., **Smith, Daniel JB**, et al. "Looking Beyond the Lamppost: Future Directions in Understanding Tropical Tree Communities." *61st Annual Meeting of the Association for Tropical Biology and Conservation*, Oaxaca, Mexico, 2025.
- Smith, Daniel JB.** "Enhancing Communication and Understanding of Theoretical Models through Dynamic Visualizations." *Modeling & Theory in Population Biology, National Institute for Theory and Mathematics in Biology*, Chicago, IL, 2025.
- Smith, Daniel JB**, et al. "Why there are so many definitions of fitness in models." *Evolution 2024*, Montreal, Canada, 2024.
- Smith, Daniel JB.** "Janzen-Connell effects and habitat-induced aggregation synergistically promote species coexistence." *Barro Colorado Island 100 Years Symposium*, Gamboa, Panama, June 2024.
- Smith, Daniel JB** and Masel, J. "The Interaction Between Exploitative and Interference Competition: Limits to R* Theory." *Evolution 2023*, Albuquerque, NM, 2023.
- Smith, Daniel JB.** "The Functional Form of Specialized Predation Affects Whether Janzen-Connell Effects Can Prevent Competitive Exclusion." *Ecological Society of America Annual Meeting*, 2021 (virtual).
- Smith, Daniel JB**, Singha, S., Banerjee, I. "Multimodel Climate Emulation of the CMIP-6 Archive." *UChicago Geoscience PhD Expo*, 2020 (virtual).
- Smith, Daniel JB** and Amarasekare, P. "Optimizations in a Variable World: On Latitudinal Trends of the Temperature Response of Fecundity." *Ecological Society of America Annual Meeting*, New Orleans, LA, 2018.
- Smith, Daniel JB** and Amarasekare, P. "On the Temperature Response of Ectotherm Fecundity." *Society of Naturalists Meeting*, Pacific Grove, CA, 2018.
- Smith, Daniel JB** and Amarasekare, P. "Towards a Mechanistic Basis of the Thermal Plasticity of Ectotherm Fecundity." *Ecological Society of America Annual Meeting*, Portland, OR, 2017.
- Smith, Daniel JB** and Amarasekare, P. "A Mechanistic Framework for Temperature Mediated Niche Partitioning in Ectotherms." *UCLA Science Poster Day*, June 2015 & *UCLA EEB Symposium*, June 2015.

TEACHING EXPERIENCE

Fundamentals of Ecology — Guest Lecturer (graduate level)

Winter 2026

The University of Texas at Austin

- Lectured on neutral theory, the niche–neutral debate, and empirical tests of niche vs. neutral theory

Introduction to Modeling in Biology — Preceptor & Guest Lecturer

Autumn 2023

The University of Arizona (upper-level undergraduate / graduate)

- Worked directly with students on computational projects in Mathematica
- Lectured on stochastic processes (Markov chain models) and the neutral theory of ecology

Population Genetics Reading Group — Group Leader

Autumn 2023

The University of Arizona

- Taught theoretical population genetics to undergraduates and graduate students
- Weekly lecture and discussion leader

Animal Behavior — Teaching Assistant (undergraduate)

Winter 2022

University of Chicago

- Mentored students on observational behavior projects
- Guest lecturer on evolutionary game theory and dynamical systems

TEACHING RESOURCES

Mathematical Concepts and Models in Ecology and Evolution (*in prep*)

Lecture chapters on fundamental models in ecology and evolutionary biology. [Chapter 2](#) | [Chapter 3](#)

Interactive Teaching / Science Communication Modules

Shiny-based interactive modules. [Ch. 2 modules](#) | [Ch. 3 modules](#)

WORKSHOPS & WORKING GROUPS

FOrest Theory and Empiricism (FORTE) Working Group at BCI 100

Summer 2024

Invited collaborator in NSF-funded tropical forest working group at the Barro Colorado Island centennial symposium.

Society for Modeling and Theory in Population Biology Working Group

Spring 2024

Invited to collaborate with mathematical biologists to plan improvements in quantitative training for undergraduate life-science students.

Conspecific Negative Density Dependence (CNDD) Working Group — Part II

Spring 2023

Invited collaborator in NSF-funded working group; contributed to one published article and two articles in prep.

Conspecific Negative Density Dependence (CNDD) Working Group — Part I

Spring 2022

Invited collaborator in NSF-funded working group; contributed to writing of two published academic articles.

AWARDS & FELLOWSHIPS

Peter O'Donnell Jr. Postdoctoral Fellowship — \$148,000 (two years)

Competitive fellowship for outstanding researchers at the interfaces of computer science, mathematics, statistics, and applications across science and engineering.

Data Science for Energy and Environmental Research (DSEER) Fellowship — \$64,000 | 2019–2021

NSF-funded data science program; developed a climate model emulator and contributed to a coding and statistics bootcamp.

NSF Graduate Research Fellowship — Honorable Mention | 2017

EEB Undergraduate Research Award, UCLA | 2016

Awarded for academic merit and senior thesis on thermal niche theory.

Whitcome Summer Undergraduate Research Fellowship — \$3,000

Formulated and analyzed the effects of temperature on ectotherm population dynamics through a theoretical model.

MENTORSHIP

Jon Staggs

Sep 2024 – Present

- UT Austin graduate student in applied mathematics / ecology
- Advising in PDE methods and manuscript preparation for ecological journals; results in prep

Tanner Muscarella

Dec 2023 – Jul 2024

- University of Arizona post-baccalaureate researcher
- Advised in applied mathematics and population genetics; results in prep

Osho Sharma

May 2023 – Jul 2024

- University of Arizona undergraduate researcher
- Advised in ecological model development; results in prep

OUTREACH

YouTube Content Creator — Life Equations

Sep 2021 – Present

Creating, animating, filming, and editing videos on theoretical concepts in ecology and evolutionary biology.
youtube.com/@LifeEquationsVideos

STEM Tutor — Chicago Tutor

Sep 2018 – Jun 2020

STEM tutoring for middle school students in Hyde Park, Chicago, through an organization supporting underprivileged youth.

UChicago Summer Camp Instructor*Jul – Aug 2018*

Supervised grades 1–6 in gardening projects and introduced grade-appropriate biological concepts.

Guest Lecturer — Restoration Ministries*Jul 2018*

Lectured on evolutionary biology and game theory to 5th–8th grade students.

ADDITIONAL WORK

Remote Tutor — Say Hello College*Jun 2024 – Present*

Tutoring high school students in SAT/ACT, AP Calculus, and AP Biology.

Freelance Academic Editor — TopEdit English Editing LLC*Oct 2020 – Sep 2021*

Line and substantive editing of academic papers by non-native English speakers.

Conservation Volunteer — American Conservation Experience (AmeriCorps)*Jun – Sep 2014*

Trail construction and maintenance, invasive species removal, and wildlife surveys in national parks and forests in Utah.

COMPUTATIONAL SKILLS

R (strong proficiency — statistical analysis, ODE simulation, stochastic programming) · **Julia** (stochastic programming and simulation) · **Mathematica** (ODE simulation, graphical analysis) · **MATLAB** · **Python** · **LaTeX**